Task B: Volumetric Water Demand Regression Report

AI-Driven Smart Irrigation Recommendation System | Precision Agriculture

1. Research Methodology & Volume Derivation

This report details volumetric water demand estimation (Water_Required_m3). Water volume equals Irrigation Depth (mm) * Field Area (605.672 ha) * 10. Setting B evaluates whether remote sensing and agro-meteorology can independently estimate volume demand.

2. Model Benchmark Summary Table

Setting	Model	Test_R2	Test_MAE_m3	Test_RMSE_m3	Test_NRMSE%	Test_MAPE%
B_Independent	RandomForest	0.6527	5659	7622	19.49	30.11
B_Independent	XGBoost	0.4597	8240	9507	24.31	53.96
B_Independent	ExtraTrees	0.8613	3751	4817	12.32	16.19
B_Independent	GradientBoosting	0.7051	4229	7024	17.96	20.97
B_Independent	ExtraTrees_Tuned	0.8611	3873	4821	12.33	19.05
A_FormulaRef	RandomForest	0.8951	2322	4189	10.71	12.81
A_FormulaRef	XGBoost	0.6001	6562	8179	20.92	47.8
A_FormulaRef	ExtraTrees	0.9096	2922	3890	9.95	12.3
A_FormulaRef	GradientBoosting	0.8804	2225	4474	11.44	12.76

3. Actual vs Predicted Volumetric Demand

